

**Advanced Econometrics (PhD in Economics) and Advanced Topics in
Econometrics (PhD in Mathematics Applied to Economics and Management)
2025/2026**

Exercise sheet 5 - Univariate time series modelling
(version 26/10/2025)

1. Which of the following autoregressive processes are stationary? [where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$]
 - (a) $Y_t = 5 - 0.2Y_{t-1} + 0.08Y_{t-2} + \varepsilon_t$;
 - (b) $\Phi(L)Y_t = \varepsilon_t$, where $\Phi(0.1 \pm 3i) = 0$, $\Phi(-1 \pm 0.1i) = 0$, $\Phi(0.9 \pm 0.4i) = 0$ and $i = \sqrt{-1}$. In this $AR(p)$ process what is the value of p ?
2. Which of the following moving average processes are invertible? [where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$]
 - (a) $Y_t = 2 + \varepsilon_t + 0.1\varepsilon_{t-1} - 0.3\varepsilon_{t-2}$.
 - (b) $Y_t = -1 + \varepsilon_t - 0.99\varepsilon_{t-1} - 0.01\varepsilon_{t-2}$.
3. Consider the stationary $AR(2)$ process $Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$, where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.
 - (a) Derive an expression for $\mu = E(Y_t)$.
 - (b) Derive an expression for $\gamma_0 = var(Y_t)$ and for $\rho_j = \gamma_j / \gamma_0$, where $\gamma_j = E[(Y_t - \mu)(Y_{t-j} - \mu)]$, ($j = 1, 2, \dots$).
 - (c) Under what conditions is Y_t stationary?
 - (d) Write down the $MA(\infty)$ representation of the $AR(2)$ process.
4. Consider the stationary $ARMA(1, 1)$ process $Y_t = c + \phi_1 Y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$, where $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.
 - (a) Derive an expression for $\mu = E(Y_t)$.
 - (b) Derive an expression for $\gamma_0 = var(Y_t)$ and for $\rho_j = \gamma_j / \gamma_0$ where $\gamma_j = E[(Y_t - \mu)(Y_{t-j} - \mu)]$, ($j = 1, 2, \dots$).
 - (c) Under what conditions is Y_t stationary? Under what conditions is Y_t invertible?
 - (d) Write down the $MA(\infty)$ representation of $ARMA(1, 1)$ process assuming that it is stationary.
 - (e) Write down the $AR(\infty)$ representation of $ARMA(1, 1)$ process assuming that it is invertible.

5. Let

$$X_t = \phi X_{t-1} + \varepsilon_t,$$

and

$$Y_t = \eta_t + \theta \eta_{t-1},$$

where ε_t and η_t are independent white noise processes with variances σ_ε^2 and σ_η^2 respectively, and $|\phi| < 1$. Let $Z_t = X_t + Y_t$.

- (a) Show that

$$Z_t = \phi Z_{t-1} + v_t,$$

where $v_t = \varepsilon_t + \eta_t + (\theta - \phi)\eta_{t-1} - \phi\theta\eta_{t-2}$.

- (b) Obtain the autocorrelation function of v_t . Given the obtained result, what is the ARMA model of Z_t ?